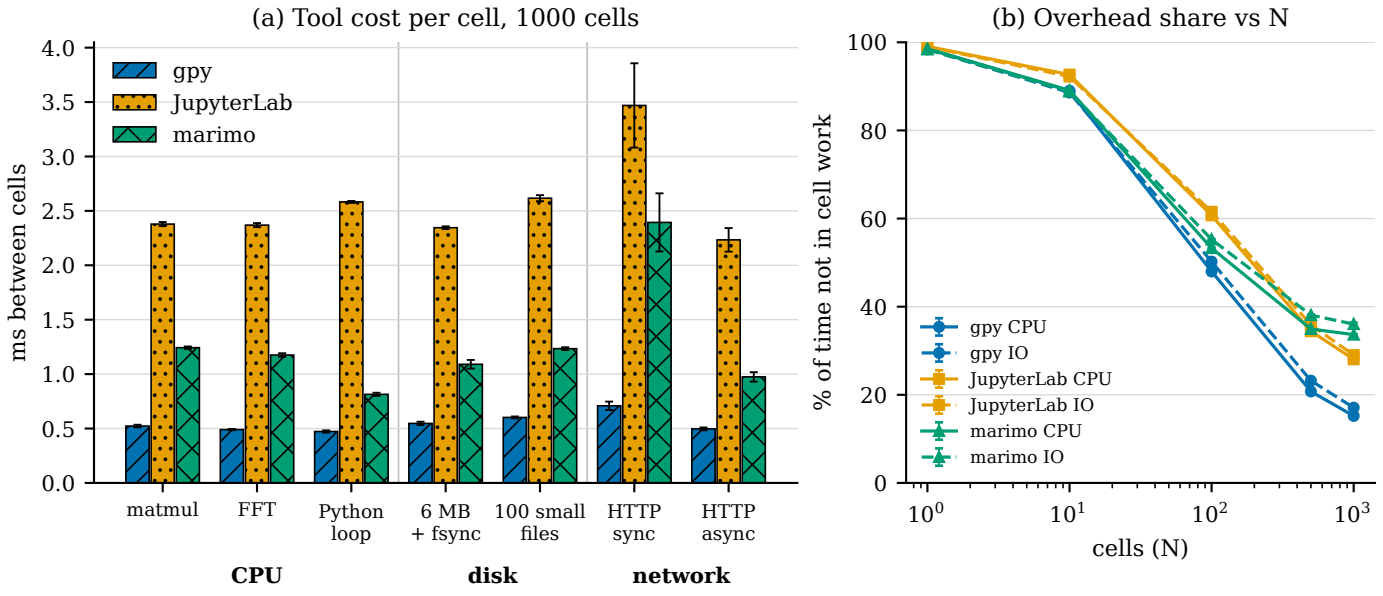

(a) Tool cost per cell, 1000 cells
ms between cells
gpy
JupyterLab
marimo
matmul
FFT
Python loop
6 MB + fsync
100 small files
HTTP sync
HTTP async
CPU
disk
network
(b) Overhead share vs N
% of time not in cell work
cells (N)
gpy CPU
gpy IO
JupyterLab CPU
JupyterLab IO
marimo CPU
marimo IO